

TECHNICAL

Engines	Unreal Engine 4 and 5 (C++, Blueprint) · Godot 4 (GDScript, C++ GDEXTension, custom engine modules) · Unity · engines written from scratch in C++ with Direct3D 11 and in Odin
Languages	C++, C, GDScript, Python, Go, TypeScript, ActionScript
Areas	Multiplayer netcode, client prediction and server-authoritative state · gameplay and game feel · audio DSP · editors and designer tooling · UI · backend services

WORK

Self-taught programmer. Self-employed indie dev, contractor and consultant since 2012.

2025–now	Rocket Rhythm and other solo titles <i>own studio, solo</i> Designer / Programmer — Godot • Rocket Rhythm, a rhythm-action arcade shooter in Godot 4.6: GDScript gameplay over a custom C++ GDEXTension for the hot paths, including instanced arena rendering and batched web VFX. Around 900 commits since April 2026, approaching a playable demo. • Wrote a custom Godot engine module in C++ for spectral analysis — FFT, onset detection, anchor beat and BPM estimation — shared by all three rhythm titles, with a beatmap editor on top. Before settling on Godot, wrote three prototype engines from scratch to test the stack, one in C++ with Direct3D 11 and two in Odin. • RhythmShock, a Super Hexagon style rhythm game in which the song's beatmap is the obstacle course — stock Godot, GDScript, web first, shipped on Wavedash. Hard Rock Hero, an endless rhythm runner for the web, in development.
2024–2026	Unannounced Project <i>undisclosed studio, contract, three person team</i> Sole Programmer • Multiplayer extraction shooter in Unreal Engine 5, built toward a playable vertical slice: client, dedicated server and backend, with gameplay written almost entirely in C++. • Multiplayer grid inventory with rotation, stacking and splitting, instant on the client and fully server authoritative. Replicated as item IDs and icon descriptors rather than full objects, to keep bandwidth down. • Character controller, weapon handling and feel, with Resident Evil 4 as the reference target: stances and aim poses, recoil, sway, spread, ADS, weapon swapping, traversal and camera framing, over a motion matching animation system.
2020–2023	Gun Jam <i>Jaw Drop Games, contract</i> Lead Developer • Implemented all game logic (C++, some BP), including the FPS player controller and several unique prototypes from scratch. • Rhythm game mechanics, developed code to correct any music desynchronization seamlessly without affecting gameplay. • Audio analysis for custom music support: low level tempo estimation and onset detection over imported audio, feeding automatic beatmap generation, plus in-game tools to build beatmaps by hand or automatically, including a training algorithm that learns from the player's inputs while playing. • Performance optimisation, tooling for artists and level designers, UI and game design support.
2016–2019	Islands of Nyne: Battle Royale <i>Define Human Studios, contract</i> Lead Gameplay Programmer • Joined a remote team as a UI developer, moved into gameplay and design, and took over as lead gameplay programmer when the previous lead left the project. Reached the contract through two years of short client work while moving from Flash and web to Unreal. • Refined core shooting mechanics and network prediction code. The game was generally praised by players for its responsive core mechanics and netcode. • Designed and built a unique ranking system for Battle Royale style games (backend and UI).
2012–2014	Fearless Fantasy <i>Enter Skies, indie team of 2</i> Programmer / Designer • Developed a custom pseudo-3D engine for Flash with AS3, later ported to a GPU accelerated sprite-based engine for desktop and mobile. Designed a gesture-based combat mechanic for turn based RPGs.
2009–2012	Nokia Maemo & Harmattan <i>Openismus GmbH</i> Software Developer • Built GUI styling tools with C and C++. Worked with designers to implement a novel mobile UI, and supported programmers in adopting those designs.

EARLIER

2005–2011	Co-author of Clearlooks , the GTK theme engine adopted as the default theme in GNOME 2.12 and used until GNOME 3, and of the initial UbuntuLooks fork behind Ubuntu's default theme. Other open source work: a text editor and a chess (PGN) viewer.
earlier	First public implementation of server side reconciliation netcode for Unreal Tournament, 2003. Extensive modding of idTech and Unreal based games. Flash games including Battle Scribes, sponsored by Kongregate. Full stack web work throughout, including a progressive web app to play Go online with a static score analyser, and backend and community tooling in Go.